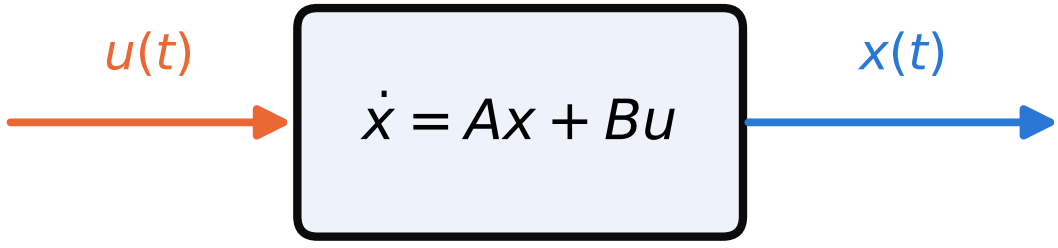
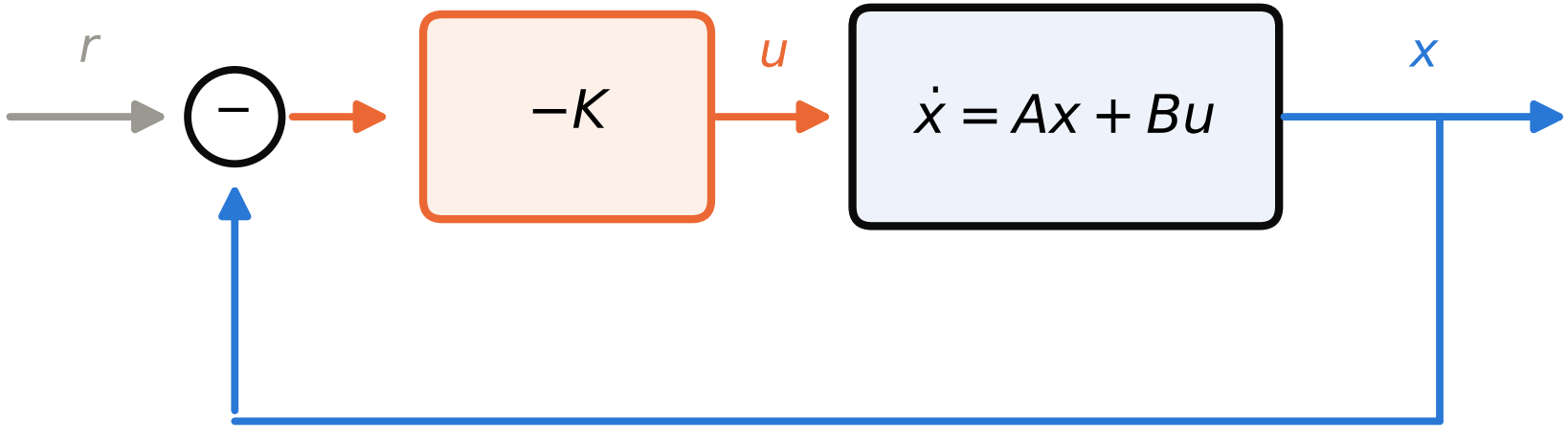
open loop — plan once, hope



closed loop — measure and react



the loop that makes it robust